Phishing Analysis: MediSure Invoice

Email Header Analysis CASE 125

AMDARI
Victor Umunna 07/10/26

Project Scope

01 Message credibility

Assess how branding and invoice details make the request appear legitimate.

02 Technical evidence

Compare sender identity, authentication headers and embedded destinations.

03 Business Exposure

Explain potential payment, account and operational risks.

04 Proposed Response

Define verification steps and a training evaluation plan.

Message Credibility

A phishing investigation was carried out by me after I contacted by a client MediSure to investigate a quarantined email. At first, when I checked the email header- I discovered some irregularities in the mail. With further forensics investigation I understood that a payment of $18,740 was demanded and it was meant to be paid in 24hrs with is typical of a phishing email.

Technical Evidence

After some close analysis I discovered some many reasons that establishes the fact that this was actually a phishing email. Some of those points includes;

Evidence in the sample	Recorded value	Meaning for review
Visible sender	billing@medequip-solutions.com	Vendor identity claim
Reply-To	medequip.accounts.recovery@gmail.com	Replies divert to another identity
Return-Path	bounce@medequip-invoices.com	Different sending domain
Authentication-Results	SPF fail / DKIM none / DMARC fail	Failed authentication

Identity mismatches support escalation alongside the message content.

Technical Evidence

Aside the SPF, DKIM, DMARC authentication fails, the reply to and the return path address being different, the domain name being different from the sender's name.There were a lot of things wrong with the mail that I discovered in terms of attachments and the timeline.

The message promises a PDF

The actual attachment is Invoice_MSH-2024-0912.html, with content type text/html.

Placeholder Attachment

Its script contains only a comment describing a simulated HTML-smuggling payload.

Time Scale

Claimed invoice due date was 7th September 2024, but the email only came two days later which was 9th September 2024. The two-day gap conflicts with the stated "more than 30 days past due" condition.

Business Exposure

Payment Diversion
A user could authorize an unverified invoice or follow untrusted payment instructions.

Account Exposure
A billing login could seek credentials. The sample does not show the destination page.

Workflow Disruption
The delivery threat could push staff to bypass approval and vendor checks.

Independent Verification
A user could authorize an unverified invoice or follow untrusted payment instructions.

Proposed Response

After the analysis were done, the next course of action taken was ascertaining if any damages had occurred, how to tackle them and also a prevention plan was set in motion.

Confirm, preserve & Investigate

After I establish that no one clicked, opened the HTML attachment, I preserved a copy of the email for further analysis and as evidence.

Contain Confirmed Impact

The credentials weren't exposed, so no need to reset them, change passwords or revoke sessions. No payment occurred, hence there was no need contacting the bank.

Assigned Accountability Owners

I helped assign clear responsibilities eg-Security owns technical controls. Finance owns payment verification. Agree actions, deadlines and evidence.

Further actions taken includes;
Blocking the domain of the sender, the sender and the URLs.
Checking proxy and DNS visits.

Recommendations

Immediate: Payment Safeguard
Use trusted supplier records and independent callbacks. Urgency must follow the normal approval process.

On going: Control Review
Review email filtering, patching and access rights. Track exceptions and close actions with evidence.

Within 30 days: Account Protection
Require MFA for email and finance accounts. Prefer phishing-resistant methods where supported.

Within 30 days: Account Protection
Provide a clear reporting route. Train staff on sender mismatches, actual link domains and unexpected files.



VICTOR UMUNNA

Victor Umunna holds a Masters degree in Cyber Security Technology from the previous Northumbria University. He has been in the IT space for over 5 years. He is currently working as a SOC Analyst as an Intern with Amdari.
He also has a wealth of experience in UI/UX, product design, graphics designs and videography as well.

ivictorumunna@gmail.com

Thanks